

Blinkit Grocery Sales Business Analysis Report

Executive Summary

This business intelligence report delivers an end-to-end operational performance analysis of Blinkit's retail infrastructure across **\$1.20M in total sales revenue**. The executive objective is to trace data flows directly from raw file cleaning stages to advanced relational database querying and final desktop business dashboards. By integrating critical store factors like location tiers, outlet size, product classes, and shelf presence metrics, this framework establishes a single source of truth allowing C-suite executives to diagnose channel health and scale logistical networks within 60 seconds.

Business Problem & Objectives

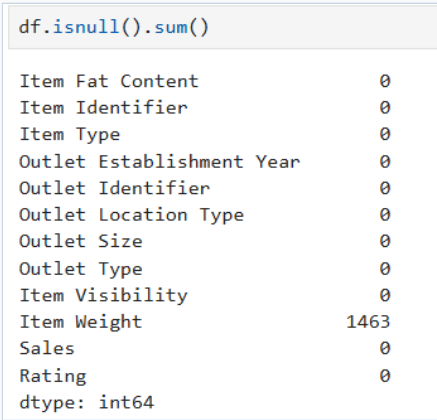
Rapidly scaling quick-commerce demands agile inventory planning and precise localized expansion strategies to avoid regional bottlenecks. The core operational targets solved by this data pipeline are:

- Analyzing transactional trends across distinct geographic location classifications (Tier 1, Tier 2, Tier 3) and outlet layout structures to guide future real estate allocation.
- Evaluating the correlation between product health traits (Fat Content) and customer item type frequencies to streamline stock turnover velocity and eliminate procurement waste.
- Isolating item shelf visibility vectors to measure their direct mathematical effect on average unit tickets.

Data Overview & Cleaning Process

The transactional matrix covers **8,523 independent records** across **12 data fields**. Baseline profiling exposed structural integrity anomalies including **1,463 missing observations** inside the Item Weight index and systemic text syntax variations across categorical fields. Data pipeline scrubbing was performed using Python Pandas:

A. Missing Value Identification & Dynamic Imputation



df.isnull().sum()	
Item Fat Content	0
Item Identifier	0
Item Type	0
Outlet Establishment Year	0
Outlet Identifier	0
Outlet Location Type	0
Outlet Size	0
Outlet Type	0
Item Visibility	0
Item Weight	1463
Sales	0
Rating	0
dtype: int64	

Screenshot 3.1: Null profiling output via df.isnull().sum()

Technical Explanation (Screenshot 3.1): Initial database checks flagged extensive missing fields restricted to the Item Weight property. To protect standard deviations and prevent modeling bias, a customized category imputation step was required.

```
df['Item Weight'] = df['Item Weight'].fillna(df.groupby('Item Type')['Item Weight'].transform('median'))

df['Item Weight'].isnull().sum()

np.int64(0)

df.isnull().sum()

Item Fat Content      0
Item Identifier      0
Item Type             0
Outlet Establishment Year 0
Outlet Identifier     0
Outlet Location Type  0
Outlet Size          0
Outlet Type          0
Item Visibility       0
Item Weight          0
Sales               0
Rating              0
dtype: int64
```

Screenshot 3.2: Category-wise median fillna() transformation and validation

Technical Explanation (Screenshot 3.2): Missing records were dynamically resolved by executing a nested `transform('median')` mapped directly to each distinct `Item Type` partition. The subsequent validation layer confirmed an absolute residual count of 0, preserving total record integrity.

B. Categorical Discrepancy Standardization

```
df['Item Fat Content'].value_counts()
```

```
Item Fat Content
Low Fat      5089
Regular     2889
LF           316
reg          117
low fat      112
Name: count, dtype: int64
```

Screenshot 3.3: Naming inconsistencies inside Item Fat Content column

Technical Explanation (Screenshot 3.3): Frequency evaluations uncovered human-entry syntax variance where equivalent attributes were logged redundantly as LF, low fat, and Low Fat, inflating distinct row calculations.

```
df['Item Fat Content'] = df['Item Fat Content'].replace({
    'LF': 'Low Fat',
    'low fat': 'Low Fat',
    'reg': 'Regular'
})
```

```
df['Item Fat Content'].value_counts()
```

```
Item Fat Content
Low Fat      5517
Regular     3006
Name: count, dtype: int64
```

Screenshot 3.4: Mapping dictionary execution via replace() and normalization

Technical Explanation (Screenshot 3.4): An explicit key-value replacement mapping matrix compressed these disparate data strings into two clean, standardized relational parameters: `Low Fat` (5,517 rows) and `Regular` (3,006 rows), neutralizing the attributes for optimized database insertion.

Key Insights & Exploratory Data Analysis (EDA)

Following data cleaning, the standardized database was migrated into PostgreSQL via an active SQLAlchemy engine connection to run heavy analytical business scripts:

1. Partitioned Window Functions (Top 3 Best Selling Items within Each Item Type):

	Item_rank bigint	Item_Type text	Item_Identifier text	Total_sales numeric
1	1	Baking Goods	FDU12	2371.01
2	2	Baking Goods	FDW60	1764.70
3	3	Baking Goods	FQJ48	1725.56
4	1	Breads	FDS59	2095.59
5	2	Breads	FDP11	1741.13
6	3	Breads	FDR23	1414.20
7	1	Breakfast	FDO37	1848.26
8	2	Breakfast	FDL13	1631.81
9	3	Breakfast	FDR37	1277.20
10	1	Canned	FDP25	1955.14
11	2	Canned	FDS13	1857.12
12	3	Canned	FDW49	1414.10
13	1	Dairy	FDW26	2002.29
14	2	Dairy	FDA15	1995.97
15	3	Dairy	FQZ26	1914.68
16	1	Frozen Foods	FDF05	2103.13
17	2	Frozen Foods	FDP28	2087.85
18	3	Frozen Foods	FDA04	2072.07
19	1	Fruits and Vegetables	FDT07	2306.90
20	2	Fruits and Vegetables	FDX31	2104.46
21	3	Fruits and Vegetables	FDY55	2058.19
22	1	Hard Drinks	DRN47	1614.79
23	2	Hard Drinks	DRK59	1408.17
24	3	Hard Drinks	DRF23	1389.62
25	1	Health and Hygiene	NQD53	1655.51

Screenshot 4.1: PostgreSQL output console showing partition rankings (Part 1)

26	2	Health and Hygiene	NCS29	1992.73
27	3	Health and Hygiene	NCM85	1581.74
28	1	Household	NCQ06	2294.71
29	2	Household	NCB31	2104.73
30	3	Household	NCS06	1835.34
31	1	Meat	FDJ55	1797.82
32	2	Meat	FDQ39	1524.18
33	3	Meat	FDI19	1458.21
34	1	Others	NCL31	1288.22
35	2	Others	NCL55	1270.62
36	3	Others	NCJ31	1207.70
37	1	Seafood	FDG33	1722.96
38	2	Seafood	FDI57	1376.24
39	3	Seafood	FDJ57	1298.61
40	1	Snack Foods	FDL58	2111.65
41	2	Snack Foods	FDT21	1988.87
42	3	Snack Foods	FDR09	1814.27
43	1	Soft Drinks	DRL01	1633.17
44	2	Soft Drinks	DRF37	1574.25
45	3	Soft Drinks	DRJ24	1484.94
46	1	Starchy Foods	FDG47	1570.85
47	2	Starchy Foods	FDI11	1520.42
48	3	Starchy Foods	FDE11	1476.34

Screenshot 4.2: PostgreSQL output console showing partition rankings (Part 2)

Technical Explanation (Screenshots 4.1 & 4.2): By implementing a partitioned rank function `ROW_NUMBER() OVER(PARTITION BY item_type ORDER BY SUM(sales) DESC)`, specific best-selling inventory assets were isolated. The console maps core channel leaders such as FDU12 within Baking Goods (\$2,371.01) and FDT07 within Fruits and Vegetables (\$2,306.90) to dictate hyper-localized fulfillment plans.

2. Combined Outlet Model & Geographic Location Tier Matrix:

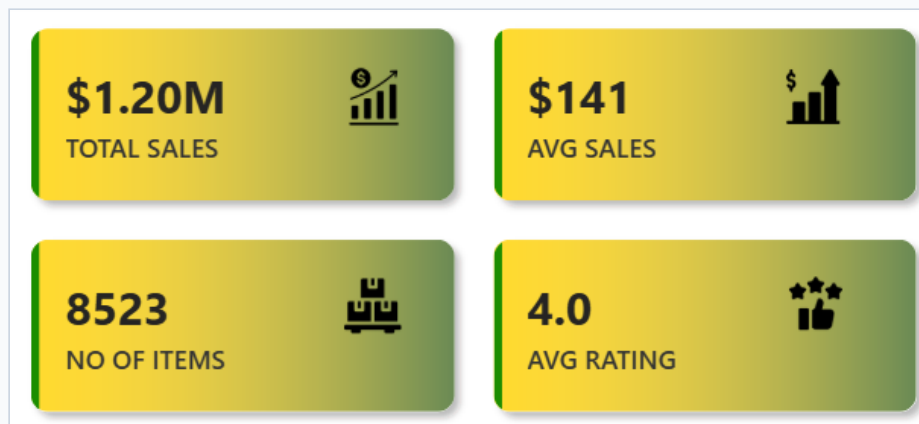
	outlet_type text	outlet_location_type text	total_sales numeric	avg_sales numeric	total_items bigint
1	Supermarket Type1	Tier 2	393150.65	141.17	2785
2	Supermarket Type1	Tier 1	262590.23	141.18	1860
3	Supermarket Type1	Tier 3	131809.02	141.43	932
4	Supermarket Type2	Tier 3	131477.78	141.68	928
5	Supermarket Type3	Tier 3	130714.67	139.80	935
6	Grocery Store	Tier 3	78131.57	140.78	555
7	Grocery Store	Tier 1	73807.58	139.79	528

Screenshot 4.3: Revenue group analysis by outlet layout and location classification

Technical Explanation (Screenshot 4.3): This multi-variable aggregation isolates the top commercial outlet sector. The **Supermarket Type1** framework deployed within **Tier 2** regions represents the primary business channel, securing **\$393,150.65** across 2,785 unique units.

Data Visualization Brief

An executive-level Power BI tracking solution was deployed to transform backend row logic into an interactive visual dashboard environment. The segmented layout and structural insights are captured below:

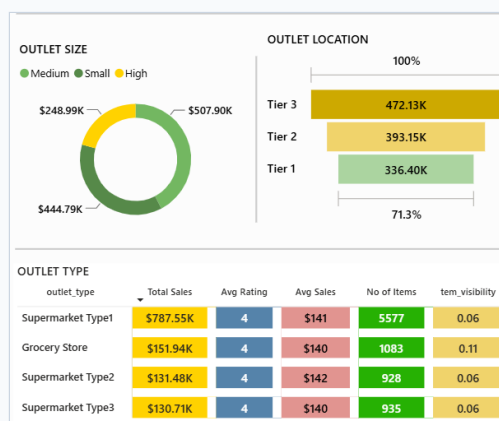


Screenshot 5.1: High-level macro corporate KPI tracking summary cards

Technical Explanation (Screenshot 5.1): Live KPI panels establish foundational operating benchmarks: Total gross sales volume stands at **\$1.20M**, average unit ticket size evaluates to **\$141**, aggregate fulfillment contains **8,523 distinct items**, and user rating averages hold strong at **4.0 out of 5.0**.

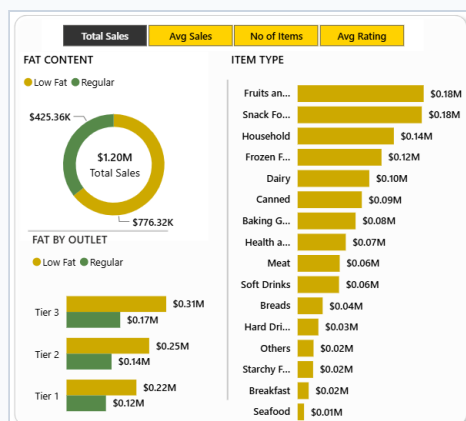


Screenshot 5.2: Master dashboard visual control panel screen layout



Screenshot 5.3: Closer look at geographical distribution funnels and outlet structures

Technical Explanation (Screenshot 5.3): The geographic funnel visual directs critical business expansions toward **Tier 3 regional markets**, which control the largest macro segment share yielding **\$472.13K in gross revenue**.



Screenshot 5.4: Close-up of consumer category weights and low-fat donut share percentages

Technical Explanation (Screenshot 5.4): The product matrix bar distribution proves that **Fruits & Vegetables** and **Snack Foods** act as twin revenue pillars (each contributing **\$0.18M**), while the corresponding fat content allocation chart confirms that health-conscious selections hold a dominant **64.6% market share**.

Business Recommendations

- **Optimize Premium Shelf Visibility Allocations:** Strategic indicators reveal an inverse relationship with average unit values (\$133.65 for high visibility vs \$143.50 for low visibility). Essential consumer goods (staples, daily groceries) retain strong inelastic demand regardless of placement. Shift premium high-visibility zones to capture high-margin impulse items (chocolates, snack lines) to boost overall average ticket sizes.
- **Scale Infrastructure Across Tier 3 Regional Corridors:** Because Tier 3 regions represent the highest gross revenue source (\$472.13K), immediate expansion budgets should prioritize rolling out advanced Supermarket Type 1 formats outside metro zones to lock in regional market capture.
- **Ensure Robust Supply Chains for Health-Conscious Inventory:** Low Fat products secure a massive 64.6% market volume share. Because average spend rates remain uniform with Regular options (\$141), logistics must establish persistent safety stock counts for Low Fat items to prevent revenue loss from out-of-stock events.
- **Coordinate Multi-Category Bundle Marketing Campaigns:** Fruits & Vegetables and Snack Foods control the highest customer basket weights (\$0.18M each). Designing bundle promotions linking these two major categories will maximize transaction throughput across primary volume drivers.